

PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

On

**Digital Forensics in Cybersecurity:
Techniques for Effective Evidence Investigation**

Submitted in partial fulfilment of the requirements for the award of degree of

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We hereby certify that the work which is being presented in the project report entitled **“Digital Forensics in Cybersecurity”** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Shikha Tuteja**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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ABSTRACT

In an era characterized by an increasing reliance on digital systems, the importance of robust cybersecurity measures has never been higher. Modern society's rapid shift toward the internet for nearly every aspect of life has been accompanied by a rise in complex cyber threats, such as data breaches, malware attacks, and unauthorized access to restricted systems. Digital forensics plays a critical role in this landscape by facilitating the identification, collection, preservation, and analysis of digital evidence required to investigate these cybercrimes. This research provides a deep dive into advanced methodologies and industry-standard tools—specifically **Autopsy** and **Wireshark**—to improve the efficiency and accuracy of digital investigations.

The study incorporates a practical dimension by simulating cyberattacks, such as systematic port scanning using **Nmap**, within a controlled virtual environment to generate realistic forensic artifacts. A significant focus is placed on the use of automation through **Python scripting** (using the **Scapy** library) to manage the challenges posed by high data volumes and the increasing complexity of digital evidence. Traditional manual investigations are often slowed down by these factors, making timely response difficult. The findings demonstrate that integrating a structured forensic framework with automated analysis significantly enhances the speed and accuracy of evidence processing while ensuring the essential legal integrity and authenticity of the data.



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1. INTRODUCTION

1.1 Background

The fast-paced evolution of technology and the widespread adoption of the internet in nearly every facet of daily life have drastically changed modern society. However, this technological progress has simultaneously introduced a complex array of cyber threats, including data breaches, phishing, malware attacks, identity theft, and unauthorized access to restricted digital systems. These cybercrimes represent severe threats to individuals, private corporations, and governments, as they can compromise confidential information and disrupt the integrity of critical infrastructures.

In response to these growing threats, digital forensics has emerged as a primary and highly effective defense mechanism. Digital forensics involves the systematic identification, collection, preservation, and analysis of digital evidence to investigate cybercrime cases. By analyzing computer, network, mobile, and storage systems, forensic experts can not only track down cybercriminals but also reconstruct the specific mechanics of how an attack occurred. This rigorous process, supported by appropriate software and standardized investigation protocols, enables cybersecurity professionals and law enforcement to collect credible evidence suitable for use in legal proceedings.

1.2 Problem Statement

Digital forensic investigators currently face a significant "data explosion" challenge. The rapid increase in the sheer volume of digital data generated daily makes conducting timely investigations increasingly difficult. Furthermore, the investigation process is often hindered by the sophisticated usage of encryption technology and anti-forensic techniques—such as data obfuscation, log manipulation, and secure data erasure—which cybercriminals utilize to obscure their digital tracks.

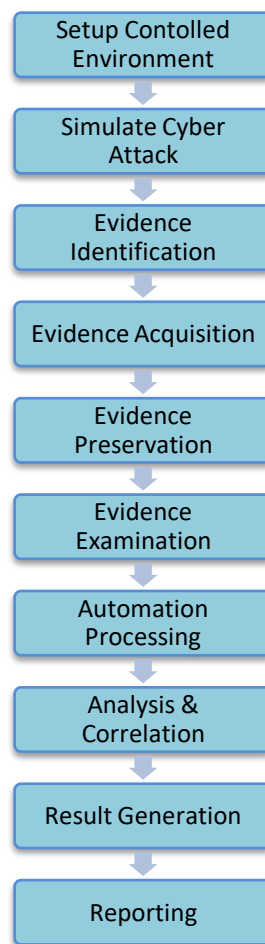
Conventional digital forensic techniques frequently fall short because they depend heavily on extensive manual human input, leading to delays and potential human error. This lack of scalability and real-time capability prevents a rapid response to ongoing threats. Consequently, there is a critical need for the development of efficient, automated procedures and structured investigation frameworks. Such automation, implemented through scripting languages like Python, is necessary to process massive datasets efficiently while ensuring the legal integrity and authenticity of the digital evidence remains uncompromised.



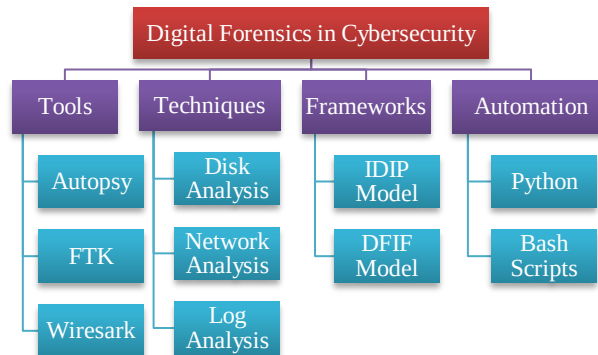
2. METHODOLOGY

The research follows a rigorous, four-stage analytical and experimental flow:

- **Experimental Environment:** To ensure safety and reproducibility, a virtual laboratory was established using VMware. This environment utilized Kali Linux as the forensic/attacking machine and Windows as the target system.
- **Forensic Investigation Process:**
 1. **Identification & Collection:** Gathering raw system logs, network captures (.pcapng), and disk images.
 2. **Preservation:** Utilizing cryptographic hashing to ensure that evidence remains in a "read-only" state and is not modified during analysis.
 3. **Examination & Analysis:** Using specialized tools to extract hidden artifacts and build a chronological timeline of events.
 4. **Reporting:** Documenting findings into structured formats such as HTML or Excel for legal and administrative review.



2. TOOLS AND TECHNOLOGIES



The following toolkit was leveraged to achieve the research objectives:

- **Wireshark:** An industry-leading network protocol analyzer used to capture and scrutinize packets to detect abnormal communication patterns.
- **Autopsy:** An open-source forensic platform used for deep-dive disk analysis, including the recovery of deleted files and the extraction of web history, cookies, and OS account details.
- **Python (Automation):** A powerful scripting environment used for automated packet processing and log analysis.
- **VMware Workstation:** The virtualization backbone providing an isolated workspace for attack emulation.



AUTOPSY
DIGITAL FORENSICS



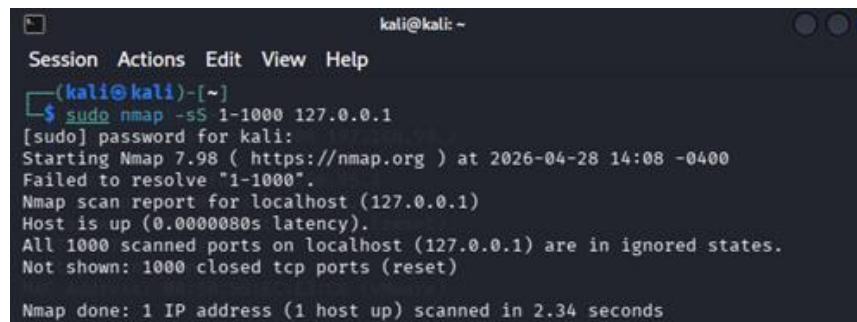
PYTHON



4. IMPLEMENTATION

4.1 Network Forensics and Automation

The **network** implementation involved executing an Nmap scan (`nmap -sS`) against the target IP to simulate connection initiation attempts. Wireshark captured this traffic, which was then filtered using `tcp.flags.syn == 1` to isolate connection requests. To optimize this, a Python script (`analyze.py`) was developed using Scapy to read the capture files directly, instantly summarizing the total packets and unique ports probed, which would take hours if done manually.



```

kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo nmap -sS 1-1000 127.0.0.1
[sudo] password for kali:
Starting Nmap ( https://nmap.org ) at 2026-04-28 14:08 -0400
Failed to resolve "1-1000".
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000000s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Nmap done: 1 IP address (1 host up) scanned in 2.34 seconds
  
```

Fig 1: Nmap port scanning command

This figure shows the execution of the Nmap command used to simulate a port scanning attack. The command sends TCP SYN packets to the first 1000 ports of the target system to identify active services.

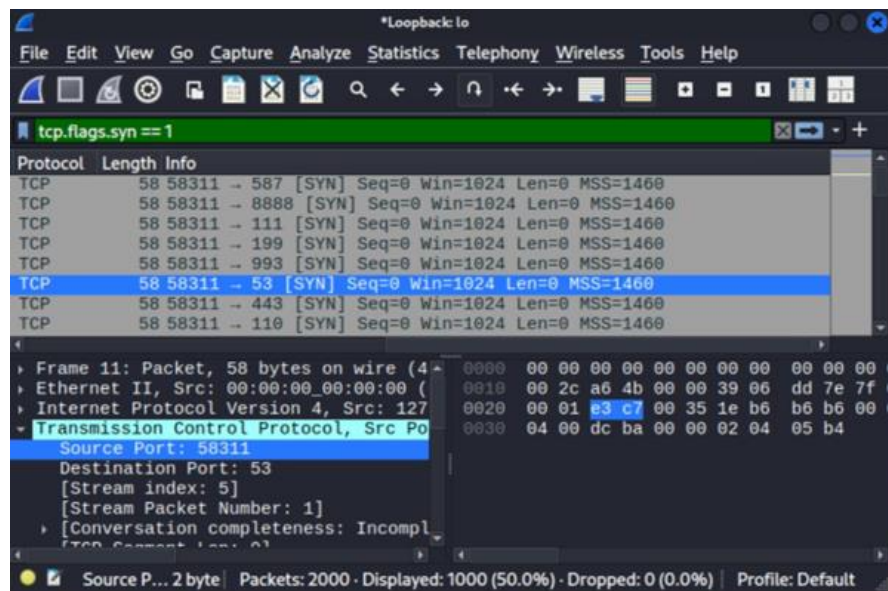


Fig 2: Wireshark capture showing SYN packets

This figure displays the Wireshark capture with the filter `tcp.flags.syn == 1` applied. Multiple SYN packets targeting different destination ports can be observed, indicating port scanning activity. The source and destination ports are visible, confirming the packet is part of a connection initiation attempt.

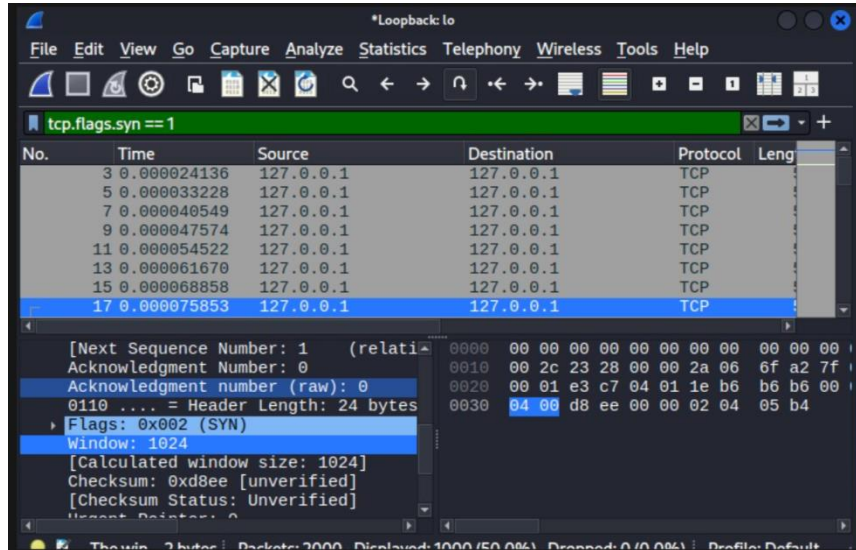


Fig 3: TCP packet details with SYN flag

This figure shows the detailed structure of a captured TCP packet. The SYN flag is set, there's no acknowledgement given back to the initiation attempt of the request, confirming that the packet is part of a connection initiation attempt which is not meant to connect a connection but scan for open connections only.

```
1 ze.py from scapy.all import *
2
3 packets = rdpcap("wireshark.pcapng")
4
5 syn_count=0
6 ports=set()
7
8 for pkt in packets:
9     if pkt.haslayer(TCP):
10         tcp = pkt[TCP]
11
12         if tcp.flags=="S":
13             syn_count+=1
14             ports.add(tcp.dport)
15             print(f"[!] SYN PACKET -> Port: {tcp.dport}")
16 print("\n---SUMMARY---")
17 print(f"Total SYN PACKETS: {syn_count}")
18 print(f"Unique Ports Scanned: {len(ports)}")
19
```

Fig 4: Python script output showing analysis

This figure presents the output of the Python script developed using Scapy. The script identifies SYN packets from the capture file and displays the destination ports being scanned.

```
(kali@kali)~$ python3 analyze.py
[!] SYN PACKET → Port: 587
[!] SYN PACKET → Port: 8888
[!] SYN PACKET → Port: 111
[!] SYN PACKET → Port: 199
[!] SYN PACKET → Port: 993
[!] SYN PACKET → Port: 53
[!] SYN PACKET → Port: 443
[!] SYN PACKET → Port: 110
[!] SYN PACKET → Port: 1025
[!] SYN PACKET → Port: 1720
[!] SYN PACKET → Port: 3389
[!] SYN PACKET → Port: 22
[!] SYN PACKET → Port: 554
[!] SYN PACKET → Port: 5900
[!] SYN PACKET → Port: 49175
[!] SYN PACKET → Port: 8701
[!] SYN PACKET → Port: 4445
[!] SYN PACKET → Port: 15004
[!] SYN PACKET → Port: 1761
[!] SYN PACKET → Port: 2007
[!] SYN PACKET → Port: 1042
[!] SYN PACKET → Port: 7070
[!] SYN PACKET → Port: 366
[!] SYN PACKET → Port: 55055
[!] SYN PACKET → Port: 8222

— SUMMARY —
Total SYN PACKETS: 1000
Unique Ports Scanned: 1000
(kali@kali)~$
```

Fig 5: Final output of total SYN packets and total unique ports scanned

This figure shows the summary generated by the Python script, including the total number of SYN packets detected and the number of unique ports scanned. This confirms the presence of a systematic port scanning attack.

4.2 Disk Forensics and Artifact Extraction

Using Autopsy, a "Local Disk" source was acquired from the Windows VM. Ingest modules were configured to automate the discovery of critical data. The software successfully categorized thousands of images, identified recently deleted files, and extracted "Shell Bags" which indicate which folders a user accessed even if they were subsequently deleted.

4.2.1 Forensic Analysis and Artifact Extraction

Following data ingestion, Autopsy performs a deep analysis of the drive, populating a structured folder system for the examiner to review.

a. Data Categories and File Views

The software categorizes discovered data into accessible views:

- **File Types:** Files are organized by extension (Images, Videos, Audio, etc.)

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
BACKGROUNDDEFAULT.jpg				2010-12-20 15:32:05 EET	2021-09-14 01:48:56 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.jpg				2011-03-27 06:04:02 EET	2021-09-14 01:48:56 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.jpg				2009-07-09 06:00:48 EET	2021-09-14 01:48:56 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.jpg				2010-12-04 17:31:22 EET	2021-09-14 01:48:56 EEST	2012-06-02 10:29:47 EEST	2012-06-02 10:29:47 EEST
BACKGROUNDDEFAULT.jpg				2009-04-04 00:34:58 EET	2021-09-14 01:48:56 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.jpg				2011-07-23 18:50:09 EET	2021-09-14 01:48:56 EEST	2012-06-02 23:58:34 EEST	2012-06-02 23:58:34 EEST
BACKGROUNDDEFAULT.jpg				2011-07-02 15:28:16 EET	2021-09-14 01:48:56 EEST	2012-06-02 23:58:34 EEST	2012-06-02 23:58:34 EEST
backgrounddefault.jpg				2011-01-12 16:30:20 EET	2021-09-14 01:48:56 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.JPG				2009-06-15 13:36:14 EET	2021-09-14 01:48:57 EEST	2012-06-02 09:41:32 EEST	2012-06-02 09:41:32 EEST
backgrounddefault.jpg				2011-03-31 04:42:05 EET	2021-09-14 01:48:57 EEST	2012-06-02 09:41:33 EEST	2012-06-02 09:41:33 EEST
BACKGROUNDDEFAULT.jpg				2011-01-14 08:34:53 EET	2021-09-14 01:48:57 EEST	2012-06-02 09:41:33 EEST	2012-06-02 09:41:33 EEST
backgroundDefault.jpg				2010-12-28 19:42:04 EET	2021-09-14 01:48:57 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.jpg				2010-08-24 15:42:32 EET	2021-09-14 01:48:57 EEST	2012-06-02 10:39:24 EEST	2012-06-02 10:39:24 EEST
BACKGROUNDDEFAULT.jpg				2010-12-04 18:29:01 EET	2021-09-14 01:48:58 EEST	2012-06-02 09:41:33 EEST	2012-06-02 09:41:33 EEST
BACKGROUNDDEFAULT.jpg				2010-09-08 23:18:57 EET	2021-09-14 01:48:59 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
backgroundDefault.jpg				2011-04-11 22:18:57 EET	2021-09-14 01:48:59 EEST	2012-06-02 23:58:35 EEST	2012-06-02 23:58:35 EEST
BACKGROUNDDEFAULT.jpg				2010-12-02 14:41:59 EET	2021-09-14 01:48:59 EEST	2012-06-02 10:49:20 EEST	2012-06-02 10:49:20 EEST
BACKGROUNDDEFAULT.jpg				2011-01-01 18:19:57 EET	2021-09-14 01:48:59 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST
BACKGROUNDDEFAULT.jpg				2010-07-15 20:23:08 EET	2021-09-14 01:48:59 EEST	2012-06-02 09:36:14 EEST	2012-06-02 09:36:14 EEST

Fig. 6: Images Details

BACKGROUNDDEFAULT.jpg			2011-01-01 18:19:57 EET	2021-09-14 01:48:59 EEST	2012-06-02 09:36:14 EEST	201
BACKGROUNDDEFAULT.jpg			2010-07-15 20:23:08 EEST	2021-09-14 01:48:59 EEST	2012-06-02 09:36:14 EEST	201
BACKGROUNDDEFAULT.jpg			2009-08-16 15:01:38 EEST	2021-09-14 01:48:59 EEST	2012-06-02 09:36:14 EEST	201

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Case		Data Source Name		File Name					
Item has no attributes with...									

Central Repository Starting Date: Not Available.

Fig. 7: Particular Image's Detail

- **Error Reporting and Messages:** A dedicated message icon provides access to status updates. These messages are essential for troubleshooting issues related to information extraction.

Module	Num	New?	Subject	Timestamp
Hash Lookup	1	●	No notable hash set.	2021/10/26 14...
Hash Lookup	1	●	No known hash set.	2021/10/26 14...
Recent Activity	1	●	Started PhysicalDrive0	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.python.proxies.GPX_Parser_Mo...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.python.proxies.module\$Android...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.centralreposit...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.keywordsearc...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.data...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.dron...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.embe...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.encr...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.filety...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.hash...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.inter...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.leapp...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.leapp...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.pictu...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.vmx...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.modules.yara...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.recentactivity...	2021/10/26 14...
Hash Lookup	1	●	Read Error: org.sleuthkit.autopsy.thunderbird...	2021/10/26 14...

Sort by: Time Total: 21 Unique: 21

Fig. 8: Messages

- **Deleted Files:** The system identifies and segregates recently deleted files from the file system for further scrutiny.
- **Metadata:** Detailed timestamps (Modified, Changed, Access, and Created times) are provided for every identified file.

b. User Activity and Web Artifacts

The examiner can reconstruct user behavior by analyzing several data artifacts:

- **Recent Activity:** Includes recently accessed documents and programs currently or previously run on the system.
- **Web History:** Comprehensive logs of web cache, cookies, downloads, and search history are extracted.

[Add Data Source](#)
[Images/Videos](#)
[Communications](#)
[Geolocation](#)
[Timeline](#)
[Discovery](#)
[Generate Report](#)
[Close Case](#)

[Keyword Lists](#)
[Keyword Search](#)

[Listing](#)

Shell Bags:

Table	Thumbnail	Summary				
Page: 1 of 1 Pages: Go to Page: Save Table as CSV						
Source File	S	C	O	Path	Key	Last
UsrClass.dat				My Computer	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-10-25
UsrClass.dat				My Computer\{D:	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-10-25
UsrClass.dat				My Computer\{D:\CT2:8	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-10-25
UsrClass.dat				My Computer\{D:\CT2:8\Disk Analysis	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-10-25
UsrClass.dat				My Computer\{D:\CT2:8\rginal_Imager_Forensics_2021...	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-10-25
UsrClass.dat				My Computer\{Z:\OSForensics Pro 8.0 Build 1000	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-09-13
UsrClass.dat				My Computer\{Z:\OSForensics Pro 8.0 Build 1000\crack	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-09-13
Shell.bag				My Computer\{Z:\AccessData Registry Viewer 2.0	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-09-13
UsrClass.dat				My Computer\{C:	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-09-13
UsrClass.dat				My Computer\{C:\ProgramData	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-09-13
UsrClass.dat				My Computer\{C:\ProgramData\Microsoft	Local Settings\Software\Microsoft\Windows\Shell\Bag\MP\U...	2021-09-13

8

Source File	S	C	O	Comment	File Path
 data_3_c103074c				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 f_0000bd				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 f_0000be				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 f_0000cc				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 f_0000cc				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 f_0000d5				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 f_0000d7				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Users/Administrat
 WelcomeScan.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/ProgramData/Micr
 Annie Sleeping 1.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Program Files (x8
 Annie Sleeping 2.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Program Files (x8
 Annie Sleeping 3.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Program Files (x8
 AnnieAtBeachDigging.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Program Files (x8
 AnnieAtBeachRunning.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Program Files (x8
 AnnieBavWhat.jpg				EXIF metadata data exists for this file.	/img_PhysicalDrive0/vol_vcl3/Program Files (x8

Fig.

-
- The screenshot shows a Windows File Explorer window with the address bar displaying 'This PC > Data Artifacts'. The left sidebar shows the 'Data Artifacts' folder selected. The main pane displays a list of subfolders and files within 'Data Artifacts':
- File Types
 - Deleted Files
 - MB File Size
 - Data Artifacts (expanded)
 - Installed Programs (37)
 - Metadata (22)
 - Opened or System Information (2)
 - Recent Documents (13)
 - Run Programs (127)
 - Shell Eggs (56)
 - USB Device Attached (2)
 - Web Bookmarks (2)
 - Web Cores (1090)
 - Web Cookies (797)
 - Web Downloads (5)
 - Web History (4008)
 - Web Search (16)
 - Analysis Results
 - Encryption Suspected (2)
 - EXIF Metadata (35)
 - Extension Mismatch Detected (3)
 - Interesting Items (1)
 - Keyword Hits (3004)
 - Single Literal Keyword Search (0)
 - Single Regular Expression Search (0)
 - Email Addresses (304)
 - IPv4 (192.168.0.0/24 + 1) + (192.168.0.0/24 + 1) + 1 (17) (10)
 - User Content Suspected (35)
 - Web Categories (5)
 - Unknown

Name	Login Name	Realm Name	Creation Time
 S-1-S-18	zyscenprofile		
 S-1-S-80-556030885-3418522549-1831038044-185329			
 S-1-S-21-3260323025-2231885063-2106927022-500			
 S-1-S-19	LocalService		
 S-1-S-21-6357229659-3247113798-1909539115-500	Administrator		2021-09-13 05:32:12 EEST
 S-1-S-80-262925252-48-4217863784-3378508180-26591			
 S-1-S-20	NetworkService		
 S-1-S-21-1235567878-628510310-3235991629-10C1			
 S-1-S-21-6357229659-3247113798-1909539115-5C1	Guest		2021-09-13 05:32:12 EEST

File	Test	File Search	File Preload	OS Account	OSa Profiles	Analysis Results	Context	Attachments	Other Resources
Basic Properties Login: Administrator Full Name: S-I-5-21-3357229699-3247113798-1909539116-500 Address: Type: Creation Date: 2021-09-13 06:02:12 EEST									
PhysicalDrive0_1 Host Details									

Fig. 12.1: OS Accounts

The screenshot shows a software interface with a tabbed menu at the top. The 'OS Account' tab is selected and active. It displays the following information:

- Login Count:** 25
- Administrator:** True
- Description:** Built-in account for administering the computer/domain
- Password Settings:** Password does not expire, Password not required
- Flag:** Normal user account
- Home Directory:** C:/Users/Administrator

Below the account details, there is a section titled 'Realm Properties' with the following information:

- Name:** Unknown
- Address:** S-1-5-21-3357229699-3247113798-1909539116
- Scope:** Local
- Confidence:** Inferred

Fig. 12.2: Particular OS Accounts Details

4.2.2. Timeline and Data Summary

To establish a chronological sequence of events, the **Timeline** tool visualizes file activity and events over specific years, months, or days. A comprehensive **Data Source Summary** provides a high-level overview of the drive, including a breakdown of file types (Documents, Executables, Images, etc.) and the overall ingest status.

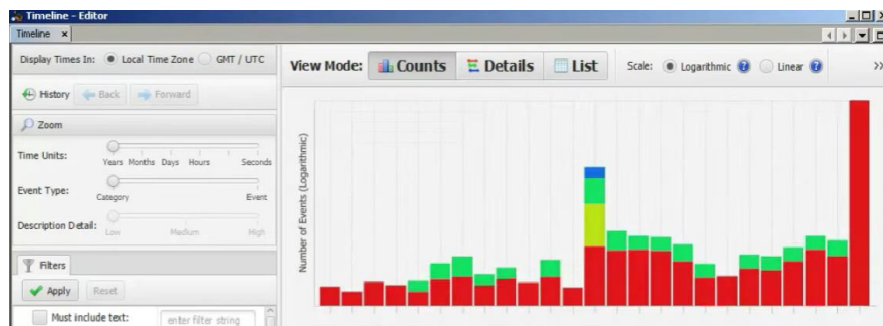


Fig. 13: Timeline

The screenshot shows the 'Data Sources Summary' window. It contains a table with the following data:

Data Source Name	Ingest Status	Type	Files	Results	Tags
PhysicalDrive0	Started	OS Drive (Windows 7 Ultimate S...	108177	15262	0

Below the table, there are tabs for 'Types', 'User Activity', 'Analysis', 'Recent Files', 'Past Cases', 'Geolocation', 'Timeline', 'Ingest History', 'Container', and 'Export'. The 'Types' tab is currently selected, showing 'Usage: OS Drive (Windows 7 Ultimate Service Pack 1)'.

Fig. 14.1: Data Source Summary

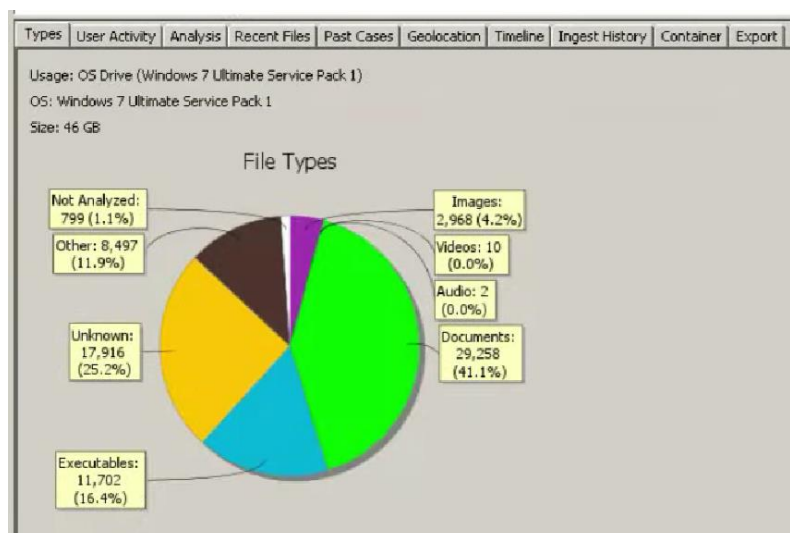


Fig. 14.2: Data Source Summary

4.2.3. Final Reporting

The final stage of the investigation involves consolidating all findings into a formal document.

- **Format Selection:** The examiner can generate reports in several formats, including **Excel**, **Text**, or **HTML**.
- **Configuration:** For this case, an **HTML report** was selected. The examiner can customize headers and footers to meet jurisdictional or organizational requirements.
- **Export:** The resulting report includes all relevant details from the investigation—including case summaries, EXIF metadata, and flagged items—ready for viewing in a standard web browser.

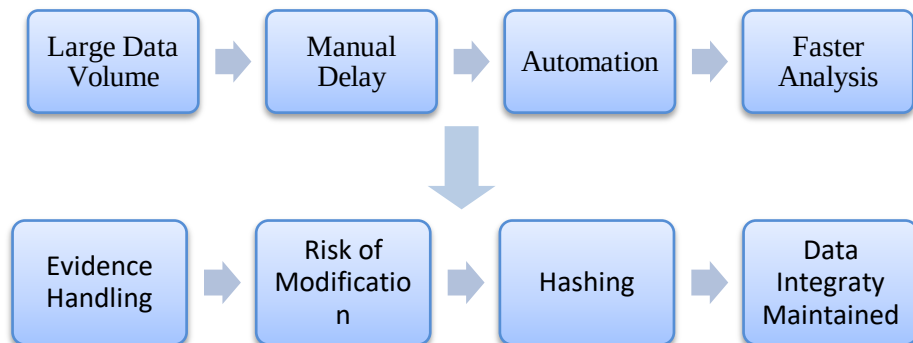
5. MAJOR FINDINGS / OUTCOMES

The experiment proved that automation and structured tools significantly enhance investigation outcomes:

Analysis Tool	Disk Analysis	Network Monitoring	Log Processing	Data Recovery
Autopsy	✓(2253 Images)	✗	✓	✓(Deleted Files)
Wireshark	✗	✓(SYN Patterns)	✓	✗
Python	✓(Custom Scripts)	✓(Scapy)	✓(Automated)	✗

Key Results:

- **Attack Detection:** The automated script identified 1000 SYN packets targeting 1000 unique ports, confirming a systematic reconnaissance attack.
- **Artifact Recovery:** Autopsy identified two suspected encrypted files and extracted 4008 web history entries, allowing for the reconstruction of user behavior.
- **Integrity:** Hashing ensured that the evidence remained authentic throughout the lifecycle.



6. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This research successfully demonstrates that while cybersecurity threats are rapidly evolving in both complexity and scale, they can be countered effectively through a multi-layered investigative approach. The study proves that the combination of a structured digital forensic framework, specialized industry software like **Autopsy** and **Wireshark**, and custom **Python automation scripts** provides a robust methodology for identifying and analyzing malicious network activity.

Key findings indicate that systematic procedures—from initial packet capture and disk imaging to the preservation of evidence through cryptographic hashing—are vital for ensuring that digital artifacts remain tamper-proof and credible for legal use. Furthermore, the integration of automation via scripting was shown to drastically reduce the time required to process vast datasets, such as identifying a 1000-port systematic scan from thousands of raw packets. Ultimately, these combined techniques are essential for modern incident response, allowing investigators to move beyond manual inspection to provide rapid, accurate, and high-integrity forensic conclusions.

6.2 Future Scope

- **AI/ML Integration:** Future work should focus on utilizing Artificial Intelligence and Machine Learning to automate the detection of abnormal network patterns. Developing intelligent systems would enable forensic tools to not only analyze past incidents but also predict and detect sophisticated attack patterns in real-time, significantly accelerating the search for signs of a breach.
- **Cloud & IoT Forensics:** As digital ecosystems shift toward decentralized storage, expanding investigation methodologies to handle cloud and Internet of Things (IoT) environments is critical. This branch of forensics must address unique challenges related to data ownership, cross-jurisdictional evidence acquisition, and the restricted storage capacity of smart devices.
- **Scalable Forensics:** Given the "data explosion" in enterprise environments, there is a pressing need for forensic techniques that can scale to handle petabyte-scale data investigations. Future research should aim to refine distributed processing techniques to ensure that even massive datasets can be analyzed without causing significant delays in the investigative timeline.

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